

The Market for Elite Status*

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Abstract

Why does competition not eliminate elite institutions that consume more resources than they produce? We model a market for elite status in the general equilibrium clubs framework of Ellickson, Grodal, Scotchmer, and Zame: elite schools sell memberships that confer status but no skill; firms run by their graduates hire in the same labor market as competently managed rivals; all memberships are competitively priced. Equilibrium compensation decomposes observably: elite managers earn a *gross* wage premium — their firms’ revenue must recoup elite tuition — but a *net* discount once education costs are subtracted, which is the price they pay for status. That discount finances the compensating differential demanded by their employees and the resources burned by elite schools. Aggregate output falls monotonically as the taste for status spreads — even though elite firms produce more output per firm, they produce less output per unit of resources, and the latter is what aggregates. Yet every such equilibrium lies in the core and is Pareto optimal: no entrant, coalition, or policy commanding lump-sum instruments can profitably displace the elite sector. Markets work; material efficiency does not follow. The welfare theorem fails only when status is *positional* — valued for its scarcity. Then the equilibrium remains unique but exhibits over-entry into elite institutions: each entrant dilutes the status of every incumbent, an economy-wide externality that membership prices cannot register. Equilibria are no longer in the core, and a tax on elite memberships raises both welfare and output. Setting the economy in motion in an overlapping-generations frame, one condition — the elite sector produces less output per unit of resources — simultaneously governs the static output decline, makes status a luxury (wealthy economies endogenously support larger elites), and makes the elite sector a drag on accumulation: growth finances the status spending that arrests it, and the economy converges to a steady state permanently poorer than its status-free counterfactual, with every generation optimizing and the path fully foreseen. Positional dilution partially immunizes against the drag.

*Working draft — not for circulation. Derivations in the appendices are machine-verified; see `analysis/static_model.py` in the project repository.

1 Introduction

[Working draft of the introduction. To be finalized last, per the project plan.]

A familiar argument holds that competitive markets discipline waste: an institution that consumes more resources than it returns should be undercut by rivals and disappear. Elite institutions appear to defy this logic. Elite universities charge extraordinary tuition while their measurable value-added is contested; the firms that preferentially hire their graduates pay premia that are difficult to attribute to productivity; and both have persisted, and indeed expanded, over decades of intense competition in product, labor, and education markets.

This paper’s answer is not that competition fails, but that it succeeds. We model elite status as what it observably is: a membership — in a school, and subsequently in the class of people who run certain firms — that is bought and sold at market prices. The natural home for such a model is the general equilibrium theory of clubs developed by Ellickson et al. (1999, 2001, 2006), in which group memberships are priced alongside ordinary commodities and agents care about the characteristics of their fellow members. Within that framework we obtain three results.

First, equilibrium compensation reproduces what compensation data show. Elite managers earn a gross wage premium over their competently trained peers, because elite firms’ revenue must recoup the tuition their managers paid; net of education costs, however, elite managers accept a strictly lower payoff — the status discount that finances the system (Corollary 1). The premium decomposes exactly: gross premium equals excess tuition minus the status discount, so visible elite pay is tuition reimbursement, not productivity. The waste is thus financed by identified parties at market prices; no friction, information asymmetry, or market power is involved. The same accounting yields a sharp possibility theorem: even *strictly less productive* elite firms survive whenever the status discount exceeds the workers’ compensating differential plus the resources burned (Proposition 1) — though in that polar case the gross premium necessarily reverses, so the empirically relevant configuration is the one in which elite firms are more productive per firm and their managers visibly better paid.

Second, as the taste for status spreads through the population, aggregate output falls monotonically (Proposition 2). What matters is not productivity per firm but output *per unit of resources consumed*: in our central configuration elite firms produce strictly more output than competent ones, look superior in any firm-level comparison, pay their managers more — and still drag aggregate output down, because the elite school system burns resources that firm accounts never show.

Third — and centrally — every such equilibrium lies in the core of the economy and is Pareto optimal (Theorem 1). No entrant offering “status without waste,” no coalition of workers and investors, and no policymaker restricted to lump-sum instruments can achieve a Pareto improvement. The market has nothing to correct: falling measured output is the equilibrium price of status consumption. The popular syllogism “markets work, therefore materially wasteful institutions cannot survive competition” commits a non sequitur, and the point of this paper is to locate it precisely. Markets work; material efficiency does not follow.

The efficiency of the baseline is not a celebration; it is a diagnosis. It locates exactly where a genuine market failure must live if there is one: not in the waste, which is priced, but in the *relativity* of status. Section 7 makes status positional — valuable because others lack it — so that each entrant into the elite dilutes the status of every incumbent. This economy-wide externality lies outside the assumptions of the clubs welfare theorems: membership prices internalize composition effects *within* a group, but eliteness is a characteristic of a class that spans all elite groups. The positional equilibrium exists, is unique, and self-limits (dilution shrinks the elite), but it is no longer Pareto optimal and no longer in the core (Propositions 3 and 4): there is over-entry into elite institutions, and a Pigouvian tax on elite memberships raises welfare *and* output (Proposition 5). The contrast between Theorem 1 and Proposition 4 is the paper’s answer to when elite waste is and is not a policy problem: absolute status is efficient consumption; positional status is a rat race.

Section 8 sets the economy in motion. Each generation inherits the resources its parents saved, plays the club economy within its lifetime, and bequeaths; the static model is the period economy and its comparative statics become the transition map. A single condition — the one already governing the static output decline — delivers two further results. Status is a *luxury* by theorem: because the elite sector’s waste is a fixed resource cost while wages scale with wealth, richer economies support larger elites (Proposition 7). And the elite sector drags on accumulation: growth finances the status spending that arrests it, and the economy converges to a steady state permanently poorer than its status-free counterfactual (Proposition 8) — with no collapse, because impoverishment itself prices the elite out of existence at low wealth. The path involves no error and no myopia: it is common knowledge, and no atomistic dynasty could or would deviate. Rational societies really do let it happen.

2 Related Literature

[Skeleton — each paragraph states the confrontation to be developed.]

Markets for status. [Becker et al. \(2005\)](#) price status as a divisible good in fixed supply traded in a hedonic market. We differ in making the supply side explicit: status here is membership in clubs — schools and firms — produced by institutions with real resource costs, priced by the membership prices of [Ellickson et al. \(1999\)](#), with a production economy attached. [Cole et al. \(1992\)](#) generate status-driven distortions from matching norms; our status is marketed, not normed, which yields prices, entry margins, and comparative statics that norms models do not deliver. [Moldovanu et al. \(2007\)](#), [Board \(2009\)](#), and [Rayo \(2013\)](#) study a single designer selling status categories or exclusivity; here the supply of status is competitive and embedded in general equilibrium.

Status and allocation. [Fershtman et al. \(1996\)](#) show occupational status concerns distort the allocation of talent in growth. Our persistence result is at the level of the *firm*: identifiable, resource-inefficient organizations survive — and even output-inferior ones can — because their control rights are bundled with status consumption. The rat-race tradition ([Akerlof 1976](#); [Hopkins and Kornienko 2004](#); [Frank 1985](#); [Hirsch 1976](#)) delivers genuine inefficiency from relative comparison; in our baseline there is no externality and hence no rat race — that distinction is the point of Theorem 1, and Section 7 shows exactly what must be added to overturn it, using the widespread-externalities equilibrium concept of [Noguchi and Zame \(2006\)](#). [Kim et al. \(2024\)](#) provide a recent quantitative treatment of education-status externalities; we provide the institutional theory.

Education and elite institutions. In [Spence \(1973\)](#) education is informative about hidden types; our elite school conveys no information — there are no hidden types — and its product is the membership itself. This places us closest to the reputation-and-selection view of schooling of [MacLeod and Urquiola \(2019\)](#) and to the evidence that elite institutions’ value operates through access and affiliation rather than skill production ([Zimmerman 2019](#); [Michelman et al. 2022](#); [Chetty et al. 2026](#); [Dale and Krueger 2002](#); [Rivera 2015](#)). We treat the zero-teaching elite school as a polar case; [Arteaga \(2018\)](#) shows content matters even at elite institutions, and nothing in our results requires the polar case (Remark 1). Prior applied uses of the clubs machinery are rare; the main exceptions are [Caucutt \(2001, 2002\)](#) on schooling with peer effects, and the firms-as-clubs analyses of [Prescott and Townsend \(2006\)](#) and [Zame \(2007\)](#).

Dynamics. The rent-seeking-and-decline tradition ([Murphy et al. 1991](#); [Baumol 1990](#); [Acemoglu 1995](#)) diverts talent to unproductive activity through distorted rewards; our drag operates instead through competitively priced status consumption, and its strength

is disciplined by the same condition that governs the static equilibrium. Where [Hassler and Rodríguez Mora \(2000\)](#) and [Cole et al. \(1992\)](#) sustain aristocratic regimes through information or norms, our elite is sustained by prices — and where the leisure-loving aristocracies of [Doepke and Zilibotti \(2008\)](#) are competed away at industrialization, ours survive precisely because their inefficiency is financed by willing status buyers rather than borne by the firms' bottom line. The memberships tradition in inequality dynamics ([Durlauf 1996](#); [Galor and Zeira 1993](#)) generates traps from productive group memberships and credit constraints; our membership is a pure positional good, and our poverty dynamics run top-down — wealth creates the elite that erodes wealth — rather than bottom-up.

3 The Framework

We adopt the general equilibrium model of groups of [Ellickson et al. \(2006\)](#), which extends [Ellickson et al. \(1999\)](#) to allow groups to produce outputs and memberships to carry acquired characteristics. We state the primitives compactly and refer the reader to the source for full detail.

3.1 Commodities, groups, and memberships

There are N divisible, publicly traded private goods. Groups are described by a finite set $\mathcal{G}$ of *group types* (π, γ, y) : a *profile* $\pi : \Omega \rightarrow \mathbb{Z}_+$ specifying how many members with each *membership characteristic* $\omega \in \Omega$ the group requires; an *organizational characteristic* $\gamma \in \Gamma$; and an *input-output vector* $y \in \mathbb{R}^N$, with negative components inputs and positive components outputs. A *membership* is a pair $m = (\omega, (\pi, \gamma, y))$ with $\pi(\omega) \geq 1$; $\mathcal{M}$ denotes the (finite) set of memberships. A *list* is a function $\ell : \mathcal{M} \rightarrow \mathbb{Z}_+$; **Lists** denotes the set of lists.

3.2 Agents

The set of agents is a nonatomic finite measure space $(A, \mathcal{F}, \lambda)$. An agent a is described by a consumption set $X_a \subset \mathbb{R}_+^N \times \mathbf{Lists}$ (closed, allowing free disposal upward in private goods, with a uniform bound on list sizes), an endowment $(e_a, 0) \in X_a$ of private goods and no memberships, and a utility function $u_a : X_a \rightarrow \mathbb{R}$, continuous and strictly increasing in private goods for each fixed list. Consumption sets encode qualification requirements: a choice (x_a, ℓ_a) is feasible only if a can fulfill the requirements of every membership in

ℓ_a , and characteristics may be *acquired* by holding other memberships (e.g., a degree qualifies its holder for a managerial role). The economy is the measurable mapping $a \mapsto (X_a, e_a, u_a)$.

3.3 States and equilibrium

A *state* $(x, \mu) : A \rightarrow \mathbb{R}^N \times \mathbb{R}^M$ assigns private goods and lists. A state is *feasible* if it is individually feasible, satisfies material balance (aggregate consumption does not exceed aggregate endowment plus the net output of the groups formed), and is *consistent*: for every group type, the aggregate choices of its various memberships are in the proportions π requires, so that agents can be matched into groups with no seat empty and no agent unmatched.

Prices are $(p, q) \in \mathbb{R}_+^N \times \mathbb{R}^M$: p for private goods, q for memberships. Membership prices may be positive (tuition), negative (wages), or zero. A *group equilibrium* is a price system (p, q) with $p \neq 0$ and a feasible state (x, μ) such that:

1. **Budget balance for group types:** for each $(\pi, \gamma, y) \in \mathcal{G}$, $\sum_{\omega \in \Omega} \pi(\omega) q(\omega, (\pi, \gamma, y)) + p \cdot y = 0$.
2. **Budget feasibility:** for almost all a , $(p, q) \cdot (x_a, \mu_a) \leq p \cdot e_a$.
3. **Optimization:** for almost all a , any choice strictly preferred to (x_a, μ_a) costs strictly more than $p \cdot e_a$.

Two results from [Ellickson et al. \(2006\)](#) are used below. Their Theorem 1: *every group equilibrium state belongs to the core, and in particular is Pareto optimal; if endowments are desirable, it belongs to the strong core and is strongly Pareto optimal*. Their Theorem 2 provides existence under group irreducibility, desirable endowments, and uniformly bounded endowments; our analysis will not need it, because we exhibit equilibria in closed form.

4 An Economy with a Market for Elite Status

4.1 Goods, group types, and memberships

There are two private goods: an input good y_1 (“resources,” the numéraire) and a consumption good y_2 with price p_2 . Membership characteristics are $\Omega = \{S, M, EM, L\}$:

student, manager, elite manager, laborer. There are four group types:

$$\begin{aligned}
g_1 &= ((S); s; (-r, 0)) && \text{serious school: educates one student at resource cost } r > 0, \\
g_2 &= ((S); s^*; (-\tau r, 0)) && \text{elite school: } \tau \geq 1, \\
g_3 &= ((M, 2L); f; (-c, \phi)) && \text{competent firm: one manager, two laborers,} \\
g_4 &= ((EM, 2L); f^*; (-\beta c, \alpha \phi)) && \text{elite firm: } \beta \geq 1, \alpha > 0.
\end{aligned}$$

The parameter τ measures the resources elite schools devote to the production of status rather than skill; β measures the excess resource use of elite-managed firms; α their output multiplier. The configuration of central interest is $\tau > 1$, $\beta \geq 1$, and α *above* one but below the resource-efficiency threshold identified in Proposition 2: elite firms out-produce competent ones firm-for-firm, yet the elite complex — schools included — consumes disproportionately more resources than it returns. The polar case $\alpha \leq 1$, in which even output-inferior elite firms survive, is characterized in Proposition 1.

The memberships are $m_1 = (S, g_1)$, $m_2 = (S, g_2)$, $m_{3.1} = (M, g_3)$, $m_{3.2} = (L, g_3)$, $m_{4.1} = (EM, g_4)$, $m_{4.2} = (L, g_4)$.

4.2 Agents

Agents form a continuum of measure one. Each is endowed with $e = (k, 0)$, $k > 0$. Agents may attend at most one school and hold at most one job. The characteristic M is acquired at a serious school and EM at an elite school: a choice with $\ell(m_{3.1}) \geq 1$ is feasible only if $\ell(m_1) \geq 1$, and $\ell(m_{4.1}) \geq 1$ only if $\ell(m_2) \geq 1$. The characteristic L requires no schooling.

Agents differ in a single taste parameter $\eta \geq 0$, distributed according to an atomless distribution function F , strictly increasing on its support $[0, \bar{\eta})$ with $\bar{\eta} \leq \infty$. Preferences are

$$u(x, \ell; \eta) = D(\ell; \eta) h(x_1, x_2), \quad h(x_1, x_2) = \sqrt{x_1 x_2}, \quad (1)$$

where D is built multiplicatively from the list: attending any school contributes a factor $\delta_s \in (0, 1)$; working as a manager a factor $\delta_m \in (0, 1)$; working as a laborer a factor $\delta_l \in (0, 1)$; holding the elite-manager membership $m_{4.1}$ contributes the additional factor $(1 + \eta)$ — this is the utility of elite status — and holding the elite-firm laborer membership $m_{4.2}$ the factor $(1 - \chi)$, $\chi \in [0, 1)$, reflecting distaste for working under elite management.

Thus, for the five undominated vocational choices,

$$D = \begin{cases} 1 & \text{leisure (empty list),} \\ \delta_s \delta_m (1 + \eta) & \text{elite manager } (\ell(m_2) = \ell(m_{4.1}) = 1), \\ \delta_s \delta_m & \text{competent manager } (\ell(m_1) = \ell(m_{3.1}) = 1), \\ \delta_l & \text{laborer, competent firm } (\ell(m_{3.2}) = 1), \\ \delta_l (1 - \chi) & \text{laborer, elite firm } (\ell(m_{4.2}) = 1). \end{cases}$$

All other feasible lists (school without subsequent employment, schooling combined with labor, double schooling) are strictly dominated at any prices arising in equilibrium; Lemma 2 in Appendix A verifies this. Note that status attaches to the *elite-manager membership*: the elite school is the gate, not the prize.

It is convenient to write

$$\mu \equiv \frac{1}{\delta_s \delta_m} > 1, \quad \lambda \equiv \frac{1}{\delta_l} > 1, \quad \lambda^e \equiv \frac{\lambda}{1 - \chi} \geq \lambda.$$

Remark 1. *The elite school teaches nothing and the elite firm wastes resources by assumption; these are polar cases chosen for transparency, not necessity. All results below depend only on the composite parameters (τ, β, α) , so a reader who prefers elite schools that teach something (but less per dollar) may set τ accordingly without change to the analysis.*

5 Equilibrium

Because h is homothetic, an agent with net income w (endowment value net of membership prices) demands $x_1 = w/2$, $x_2 = w/(2p_2)$ and attains indirect utility $Dw/(2\sqrt{p_2})$. All occupational comparisons are therefore rankings of $D \cdot w$, independent of p_2 .

5.1 Characterization

Occupational indifference for the (atomless) population pins down net incomes. Since leisure yields $D \cdot w = k$, participation of laborers and competent managers requires

$$w_M = \mu k, \quad w_L = \lambda k, \quad w_{L^*} = \lambda^e k, \quad (2)$$

and the elite-manager income satisfies, for a *marginal* type η^* ,

$$w_E = \frac{\mu k}{1 + \eta^*}, \quad (3)$$

with all types $\eta > \eta^*$ strictly preferring the elite path and all $\eta < \eta^*$ strictly avoiding it. Membership prices follow from the definition of net income ($w = k - \text{fees} + \text{wages}$) and budget balance of the schools:

$$q_1 = r, \quad q_2 = \tau r, \quad q_{3.1} = k - r - \mu k, \quad q_{3.2} = k - \lambda k, \quad q_{4.1} = k - \tau r - w_E, \quad q_{4.2} = k - \lambda^e k. \quad (4)$$

Budget balance for competent firms determines the output price,

$$p_2 \phi = c + r + (\mu + 2\lambda - 3)k > 0, \quad (5)$$

and budget balance for elite firms determines the cutoff. Define

$$Z \equiv \alpha[c + r + (\mu + 2\lambda - 3)k] - \beta c - \tau r + (3 - 2\lambda^e)k. \quad (6)$$

Then the elite-firm budget balances if and only if $w_E = Z$, i.e.,

$$1 + \eta^* = \frac{\mu k}{Z}. \quad (7)$$

The measure of elite managers is $m^* = 1 - F(\eta^*)$; consistency requires the elite sector to employ $2m^*$ laborers and the competent sector, of measure n firms, to employ n managers and $2n$ laborers. Market clearing in the input good,

$$\underbrace{\sigma_0 \frac{k}{2} + m^* \frac{w_E}{2} + 2m^* \frac{w_{L^*}}{2} + n \frac{w_M}{2} + 2n \frac{w_L}{2}}_{\text{consumption demand}} + \underbrace{m^*(\tau r + \beta c) + n(r + c)}_{\text{resource use}} = k, \quad (8)$$

with $\sigma_0 = 1 - 3m^* - 3n$ the leisure share, determines n linearly; clearing of the consumption good follows by Walras' law (Appendix A).

Assumption 1 (Interior configuration). *Parameters are such that (i) $0 < Z < \mu k$; (ii) $F(\eta^*) < 1$; (iii) the solution n of (8) is strictly positive; and (iv) $\sigma_0 \geq 0$.*

Part (i) puts the cutoff in the interior: if $Z \leq 0$, elite firms cannot break even even with unpaid managers and the elite sector is empty; if $Z \geq \mu k$, every agent joins it. Parts (ii)–(iv) ensure all sectors coexist. The parameterization of Section 6 satisfies all four.

Lemma 1. *Under Assumption 1, the prices (4)–(5) and the state in which agents with $\eta > \eta^*$ sort into elite management, measures n , $2n$, and $2m^*$ of agents with $\eta < \eta^*$ sort into competent management, competent labor, and elite labor respectively, and the remainder choose leisure, constitute a group equilibrium. Every group equilibrium in which all six memberships are chosen by a positive measure of agents has these prices, and its cutoff, sector sizes, and allocation are given by (3)–(8).*

The proof (Appendix A) verifies budget balance, budget feasibility, consistency, and — using Lemma 2 — optimization over every feasible list.

Two features deserve notice. First, elite managers *accept a status discount* in net terms: $w_E < w_M$ in every interior equilibrium, with the discount $\mu k \eta^* / (1 + \eta^*)$ increasing in the equilibrium cutoff. In membership-price terms, the elite *career bundle* is always the more expensive one: $q_2 + q_{4.1} > q_1 + q_{3.1}$, tuition and job taken together. Whether the job taken alone pays more — whether observed elite *salaries* exceed competent ones — is a separate question, settled by Corollary 1 below. Second, laborers in elite firms are paid a compensating differential, $w_{L^*} > w_L$: as Ellickson et al. (2006, p. 151) anticipate, firms offering uncongenial conditions must pay more.

5.2 Gross premia, net discounts: what compensation data show

The membership prices (4) separate what an elite manager is *paid* from what an elite career *nets*. The gross wage of an elite manager is $-q_{4.1} = w_E - k + \tau r$; of a competent manager, $-q_{3.1} = w_M - k + r$.

Corollary 1 (Compensation decomposition). *In any interior equilibrium,*

$$\underbrace{(-q_{4.1}) - (-q_{3.1})}_{\text{gross wage premium}} = \underbrace{(\tau - 1)r}_{\text{excess tuition}} - \underbrace{(w_M - w_E)}_{\text{status discount}}. \quad (9)$$

Hence: (i) elite managers earn a gross wage premium if and only if the status discount is smaller than the excess tuition; (ii) a gross premium is possible only if $\alpha > 1$ (whenever $\chi > 0$ or $\beta > 1$), since the equilibrium identity $p_2 \phi(1 - \alpha) = (w_M - w_E) - 2(w_{L^*} - w_L) - (\beta - 1)c - (\tau - 1)r$ then forces $\alpha > 1$; (iii) in the polar case $\alpha < 1$ of Proposition 1, elite managers are paid less in gross terms as well.

Corollary 1 is the model’s answer to the most immediate empirical objection. Elite managers in the data are conspicuously well paid; in the model they are too, in the configuration $1 < \alpha < (\beta c + \tau r) / (c + r)$: their firms generate more revenue per firm

— enough to recoup the elite tuition embedded in their wages — and every firm-level comparison flatters them. What the data do not show, and the model prices explicitly, is the net position: subtract the cost of the elite education that the wage reimburses, and the elite career pays strictly less than the competent one. The visible premium is tuition reimbursement, not productivity — and in the limit of a vanishing status discount ($\eta^* \rightarrow 0$) it is *exactly* tuition reimbursement. The willingness to accept that worse net bargain is the status payment on which the entire elite complex is financed.

5.3 When do even unproductive elite firms survive?

Proposition 1 (Coexistence). *Under Assumption 1, an equilibrium with an active elite sector of strictly lower output productivity ($\alpha < 1$) exists if and only if, at the equilibrium cutoff η^* ,*

$$\underbrace{(w_M - w_E)}_{\text{status discount}} > \underbrace{2(w_{L^*} - w_L)}_{\text{compensating differential}} + \underbrace{(\beta - 1)c + (\tau - 1)r}_{\text{resources burned}}. \quad (10)$$

The economics is an accounting identity made competitive: the elite firm's revenue shortfall relative to a competent firm must be financed within the group, and the only willing financier is the manager who values the membership itself. The status discount conceded by the marginal elite manager must cover both the premium demanded by the two laborers who would rather work elsewhere and the resources burned by the elite school and the elite firm's own inefficiency. No cross-subsidy from outside the group is possible — budget balance for group types forbids it — so the entire institution of wasteful eliteness is self-financing, at market prices, out of the demand for status.

5.4 The spread of status preferences and the decline of output

Aggregate output of the consumption good is $Y = \phi(n + \alpha m^*)$. Consider a first-order stochastically dominating shift of the taste distribution F — more people care more about status. By (7) the cutoff η^* depends only on prices and technology, *not* on F ; hence the elite sector expands mechanically, $m^* = 1 - F(\eta^*)$ rising with the shift.

Proposition 2 (Efficient decline). *Under Assumption 1, a FOSD shift of F strictly increases the mass of elite managers and firms and strictly decreases aggregate output if and only if*

$$\frac{\alpha\phi}{\beta c + \tau r} < \frac{\phi}{c + r}, \quad (11)$$

i.e., if and only if the elite sector produces less output per unit of resources than the competent sector. In particular, output declines whenever $\alpha \leq 1$ and $(\tau-1)r + (\beta-1)c > 0$; but it also declines for a range of $\alpha > 1$.

Condition (11) is strictly weaker than $\alpha < 1$: an elite sector whose firms produce *more* output than competent firms still drags aggregate output down if its schools and firms consume disproportionately more resources. Measured productivity statistics will register this as declining aggregate TFP.

5.5 The market cannot correct it

Theorem 1 (No market correction). *Every equilibrium of Lemma 1 belongs to the core of the economy and is strongly Pareto optimal. Consequently: (i) no feasible reallocation — including any that shuts the elite sector, redirects its resources, or introduces any group type in $\mathcal{G}$ in any proportion — makes almost all agents better off without making a positive measure worse off; (ii) no coalition of agents, of any size, can profitably secede with any configuration of schools and firms; (iii) the conclusion is unchanged as the taste for status spreads and output falls per Proposition 2.*

Proof. The state is a group equilibrium (Lemma 1), so membership in the core and Pareto optimality follow from Theorem 1 of Ellickson et al. (2006). Strong Pareto optimality follows from the same theorem once endowments are verified to be desirable, which is done in Appendix B. $\square$

Theorem 1 is the paper’s central claim, and it is worth stating what it does and does not say. It does not say the elite sector is harmless: output falls, resources are burned, and the burn is real. It says the situation is not a market failure. The waste is voluntarily financed by the people who consume the status it produces, at prices that fully internalize its cost within each group. Competition has no lever: an entrant offering “elite status without the waste” would have to offer the elite-manager membership at its equilibrium price, and the equilibrium already prices every potential group type in $\mathcal{G}$. The familiar inference — markets discipline waste, elite waste persists, therefore markets must be impeded — fails at its first step. Markets discipline *unpriced* waste. Status is priced.

Three caveats delimit the result. First, the feasible set takes the group-type list $\mathcal{G}$ as given; whether a status-conferring group type without resource burn could exist is a question about what makes status credible. Section 7 answers it: when status derives from scarcity, entry self-dilutes, and exclusivity needs no artificial protection. Second, if

status is valued *because others lack it*, utility depends on the economy-wide pattern of memberships — an externality outside the present framework, under which the welfare theorem genuinely fails; that is the subject of Section 7. Third, welfare here is welfarist: status utility counts. Section 9 returns to this.

6 Numerical Example

Let $k = 2$, $r = c = 1$, $\tau = 4$, $\beta = 1.25$, $\alpha = 1.25$, $\phi = 6$, $\delta_s = \delta_m = \frac{1}{2}$ (so $\mu = 4$), $\delta_l = \frac{1}{2}$ (so $\lambda = 2$), $\chi = 0.2$ (so $\lambda^e = 2.5$), and let η be exponentially distributed with mean $s = 0.1$. Elite schools cost four times what serious schools cost. Elite firms are competent firms scaled up by exactly 25% — both input and output — so no firm-level productivity comparison distinguishes them; the entire drain is the school system's burn, $(\tau - 1)r = 3$ per elite manager trained. The resource condition (11) holds with room to spare: $\alpha(c + r) = 2.5 < \beta c + \tau r = 5.25$.

Then $p_2 = 2$, $w_E = Z = 5.75$, $\eta^* = 9/23 \approx 0.391$, and:

	elite manager	competent manager	laborer (elite firm)	laborer (comp. firm)
gross wage $-q$	7.75	7	3	2
tuition paid	4	1	—	—
net income w	5.75	8	5	4

The example displays Corollary 1 in the empirically relevant regime: elite managers earn a visible salary premium of 10.7% (7.75 against 7), which is exactly excess tuition minus the status discount ($3 - 2.25 = 0.75$), while their net position is 28% *below* their competent peers' (5.75 against 8). The elite sector employs $3m^* \approx 6.0\%$ of the population ($m^* \approx 0.020$), the competent sector $3n \approx 34.2\%$ ($n \approx 0.114$), and $\sigma_0 \approx 59.8\%$ take leisure. Aggregate output is $Y \approx 0.834$ and the resources burned are $m^*[(\beta - 1)c + (\tau - 1)r] \approx 0.065$. For the polar regime of Proposition 1, set $\tau = 2$, $\alpha = 0.9$, $s = 0.3$: coexistence condition (10) holds ($4.45 > 2 + 1.25$) and, per Corollary 1(iii), elite managers are then paid less in gross terms as well (3.55 against 7).

Figure 1 traces the equilibrium as the mean s of the taste distribution rises — a FOSD shift. The elite sector expands, the competent sector contracts, and output falls monotonically, while by Theorem 1 every point on the path is in the core.

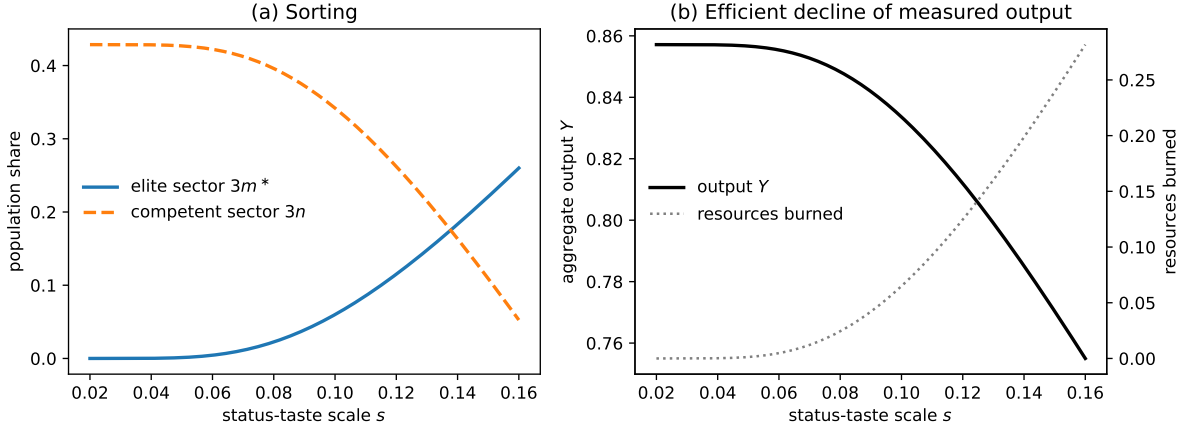


Figure 1: Comparative statics in the status-taste scale s ($\eta \sim \text{Exponential}$, mean s). Panel (a): population shares of the elite and competent sectors. Panel (b): aggregate output falls and burned resources rise as status taste spreads; every equilibrium shown is Pareto optimal (Theorem 1). Parameters as in Section 6.

7 Positional Status and the Failure of the Welfare Theorem

Theorem 1 rests on status being an *absolute* amenity: the elite-manager membership yields utility $(1 + \eta)$ regardless of how many others hold it. But much of what makes status status is scarcity (Hirsch, 1976; Frank, 1985): a credential everyone holds confers distinction on no one. This section makes status positional and shows that everything in Theorem 1 breaks — selectively, and in an instructive way.

7.1 Positional preferences and equilibrium

Let M denote the aggregate measure of agents holding the elite-manager membership $m_{4,1}$, and replace the status factor $(1 + \eta)$ in the utility specification of Section 4 with

$$1 + \eta v(M), \quad (12)$$

where $v : [0, 1] \rightarrow (0, 1]$ is continuous, strictly decreasing, with $v(0) = 1$: the value of eliteness falls as eliteness spreads. Each agent is negligible and takes M as given. Because utility now depends on *other agents'* memberships, preferences violate the assumptions of Ellickson et al. (2006): this is a widespread externality in the sense of Noguchi and Zame (2006). We accordingly define a *positional group equilibrium* as prices (p, q) and

a feasible state (x, μ) satisfying the three conditions of Section 3, where optimization is with respect to preferences (12) evaluated at the aggregate elite mass M induced by the state itself.

Crucially, the externality operates at the level of the *class* of elites — the holders of $m_{4.1}$ across all elite firms — not at the level of any single group. Membership prices internalize composition effects within a group; they have no instrument that spans groups. This is precisely the gap in the price system that the welfare theorem needs closed, and it is why the failure below is not a defect of the clubs framework but a discovery made visible by it.

7.2 Existence, uniqueness, and self-limiting eliteness

The wages (2), the output price (5), and the elite-firm break-even income $w_E = Z$ are untouched by the externality. The cutoff condition becomes $(1 + \eta^* v(M)) \delta_s \delta_m w_E = k$, i.e.,

$$\eta^*(M) = \frac{\eta_0^*}{v(M)}, \quad (13)$$

where $\eta_0^* = \mu k / Z - 1$ is the cutoff of Section 5. Consistency of the conjectured mass with sorting requires M to be a fixed point of

$$\Phi(M) \equiv 1 - F\left(\frac{\eta_0^*}{v(M)}\right). \quad (14)$$

Proposition 3 (Positional equilibrium). *Under Assumption 1 (applied at $M = M^*$), a positional group equilibrium exists and its elite mass is unique: Φ is continuous and strictly decreasing on $[0, 1]$, so it has exactly one fixed point M^* . Moreover: (i) $M^* < M_0 \equiv 1 - F(\eta_0^*)$ — dilution self-limits the elite; (ii) a FOSD shift of F raises M^* , but by strictly less than it raises M_0 — positional concerns dampen the expansion of Proposition 2; (iii) all prices, and the equilibrium values of p_2 , w_E , n , and Y given M^* , are as in Section 5.*

Part (i) already delivers a result the baseline cannot: eliteness is self-protecting. Entry of new elite institutions dilutes the very good they sell, so the market itself polices exclusivity — no entrant can arbitrage status, because relative position cannot be replicated by replication.

7.3 Failure of optimality and of core equivalence

Proposition 4 (Inefficiency). *If v is strictly decreasing, the positional group equilibrium is not Pareto optimal, and its state does not belong to the core.*

The proof (Appendix C) is constructive and exploits the equilibrium’s own structure. Every non-elite agent — laborers, competent managers, leisure-takers — is at reservation utility and indifferent among their activities; so is every marginal elite. The planner therefore shuts a small mass dM of elite triples, sends their members to reservation activities at unchanged utility, and opens competent triples in the ratio that restores material balance in both goods — a reallocation that is *exactly* feasible because both adjustment directions have zero value at equilibrium prices. Nothing real has been given up by anyone. But $v(M)$ rises, so every inframarginal elite is strictly better off: a Pareto improvement, executable by the grand coalition, so the state is blocked and core equivalence fails. The contrast with Theorem 1 is exact: there, shutting elite firms had to take status away from people who had paid its price; here, shrinking the elite *creates* status out of scarcity.

7.4 Over-entry and the membership tax

At the laissez-faire equilibrium the marginal entrant weighs the private value of elite membership against its price, but ignores the dilution imposed on every inframarginal elite. The planner’s cutoff condition (Appendix C) equates the marginal type’s surplus to the aggregate dilution wedge

$$\Omega = |v'(M)| \frac{w_E}{1 + \eta^* v(M)} \int_{\eta \geq \eta^*} \eta dF(\eta), \quad (15)$$

expressed in units of the marginal entrant’s income. Since the equilibrium sets the marginal surplus to zero while $\Omega > 0$, there is over-entry.

Proposition 5 (Membership tax). *Let a tax t be levied on the elite-manager membership, with revenue rebated as a uniform lump sum. (i) The positional equilibrium remains unique for small t , with $M(t)$ strictly decreasing. (ii) Utilitarian welfare is strictly increasing in t at $t = 0$: small membership taxes are Pareto-improvable improvements over laissez-faire in the sense of Proposition 4, and the first-order Pigouvian level is $t^* = \Omega$. (iii) If, in addition, condition (11) holds, aggregate output also rises with the tax.*

Part (iii) deserves emphasis because it reverses the baseline’s policy content. Under absolute status, taxing the elite sector reduces welfare (it undoes priced, efficient consumption). Under positional status, the tax raises measured output *and* welfare simultaneously

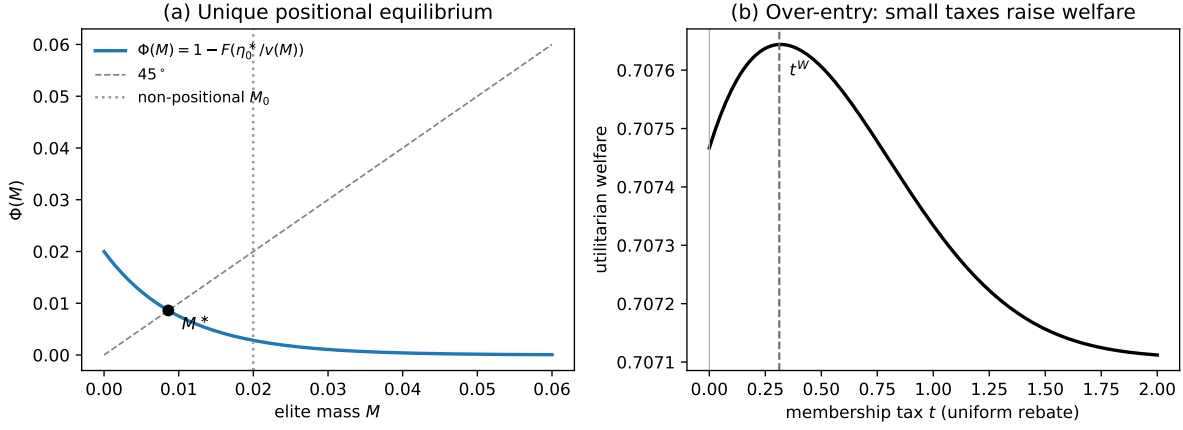


Figure 2: The positional equilibrium. Panel (a): the map Φ of (14) is strictly decreasing, giving a unique elite mass M^* , strictly below the non-positional mass M_0 (dotted). Panel (b): utilitarian welfare as a function of the membership tax t (uniform lump-sum rebate); welfare rises for all small taxes — the signature of over-entry — and peaks at t^W . Parameters as in Section 6 with $v(M) = 1/(1 + 25M)$.

— there is no equity–efficiency trade-off, because the taxed activity was destroying value at the margin in both senses. The practical diagnostic offered by the theory is thus a question about preferences, not about waste: *is elite status valued in itself, or valued because others lack it?* Only in the second case is there anything for policy to correct.

7.5 Numerical continuation

Continuing the example of Section 6 with $v(M) = 1/(1 + \psi M)$, $\psi = 25$: the elite mass falls from $M_0 \approx 0.0200$ to $M^* \approx 0.0086$ ($v(M^*) \approx 0.82$) — dilution cuts the elite by more than half — and output rises to $Y \approx 0.847$. The same FOSD taste shift that raises the baseline elite mass by 0.0085 raises the positional mass by only 0.0023. Utilitarian welfare is strictly increasing in the membership tax at $t = 0$; the welfare-maximizing tax in the example is $t^W \approx 0.31$ (about 5.4% of the elite manager’s net income, and close to the first-order Pigouvian level $\Omega \approx 0.35$), under which the elite mass falls to 0.0050 and output rises further. Figure 2 displays the fixed point and the welfare curve. All computations are verified in `analysis/positional_model.py`.

8 Dynamics: Bequests, Growth, and Rational Impoverishment

The static analysis holds wealth fixed. This section lets the economy accumulate: each generation inherits the resources its parents saved, plays the club economy of Sections 4–7 within its lifetime, and bequeaths. Three questions organize the analysis. Does the market for elite status expand or contract as an economy grows? Does the elite sector’s resource burn feed back into accumulation? And — the question any reader of Section 5 will ask — if status spending erodes the wealth of future generations, do rational, forward-looking agents really let it happen?

8.1 The overlapping-generations economy

Time is discrete, $t = 0, 1, 2, \dots$. Each period a measure-one generation is born, inherits per-capita resources k_t (the input good), and lives for one productive period within which the schooling-then-employment sequence of the static model unfolds (memberships are dated within the period, as in [Ellickson et al. 2006](#)). Preferences extend (1) with a warm-glow bequest made in the *produced* good — output invested today is the resource base of the next generation:

$$u = D(\ell; \eta) x_1^{\gamma/2} x_2^{\gamma/2} b^{1-\gamma}, \quad \gamma \in (0, 1), \quad (16)$$

so an agent with net income w spends $(\gamma/2)w$ on each consumption good and $(1 - \gamma)w$ on the bequest, and attains indirect utility $D w \Psi(p_2)$ — *still linear in net income*. Every occupational-indifference condition, membership price, the output price (5), the cutoff (7), and the positional fixed point (14) therefore carry over verbatim at wealth k_t ; only the competent-sector scale n adjusts for the saving margin, and $\gamma = 1$ recovers the static model exactly (Lemma 3 in Appendix D). Invested output converts into next-period resources at a rate $\Theta > 0$ (a linear investment technology), and children inherit the economy-wide average bequest:

$$k_{t+1} = G(k_t) \equiv \Theta (1 - \gamma) \frac{W(k_t)}{p_2(k_t)}, \quad (17)$$

where $W(k)$ is aggregate net income — endowment plus the value added of the schools and firms that form at wealth k . Uniform inheritance abstracts from within-generation dynasty heterogeneity; the Galor–Zeira-style interaction of inherited inequality with elite membership is an important extension we defer.

Proposition 6 (The transition map). *Under (16): (i) the within-period allocation at wealth k_t is the club equilibrium of Section 5 (or Section 7) at k_t ; (ii) G is continuous, with $G'(0^+) = \Theta(1 - \gamma)\phi/(c + r)$ and $\lim_{k \rightarrow \infty} G(k) < \infty$ (diminishing returns operate through the price of investment, $p_2(k)$, which rises with wealth); (iii) if $\Theta(1 - \gamma)\phi > c + r$, the origin repels and a stable interior steady state exists. Along any equilibrium path, no agent's optimization requires forecasting: bequest utility is over the gift itself, and each period's prices depend only on k_t .*

Part (iii)'s final sentence is worth dwelling on, because it answers the “do they really let it happen?” question in a form stronger than rational expectations. The path (k_t) generated by (17) is common knowledge, and every generation can compute the decline it participates in. No dynasty would behave differently with perfect foresight: each is atomistic, its bequest cannot move the aggregate, and its within-period choices are optimal at the prices it faces. The impoverishment we exhibit below is not a mistake, a bubble, or a failure of anticipation. It is what optimizing behavior looks like when status is among the things people want.

8.2 Status is a luxury — by theorem, not assumption

Proposition 7 (Luxury). *The elite cutoff $\eta_0^*(k) = \mu k/Z(k) - 1$ is strictly decreasing in k — and hence the elite mass $m^*(k)$ strictly increasing in k wherever the elite sector is viable — if and only if $\alpha(c + r) < \beta c + \tau r$: precisely the resource-productivity condition (11) of Proposition 2.*

The mechanism is instructive. The elite sector's excess costs — the school's burn $(\tau - 1)r$, the firm's burn $(\beta - 1)c$ — are *fixed* resource costs, while every wage in the economy scales with wealth. As k grows, the burn shrinks relative to the status discount that finances it, so the marginal entrant needs less taste for status to justify entry: rich societies buy more eliteness not because their preferences differ but because they can better afford the waste. Below a viability threshold ($Z(k) \leq 0$) the elite sector cannot break even at any taste level: poor economies have no market for wasteful status. No nonhomotheticity is assumed anywhere; the luxury property is produced by the same condition that makes the elite sector a drag on output.

8.3 The status drag and the spread of taste

Proposition 8 (Status drag). *At any wealth k at which the elite sector is active, $\text{sign } \partial G/\partial m^* = \text{sign}[\alpha(c + r) - (\beta c + \tau r)]$. Under condition (11), therefore: (i) the transition map lies*

strictly below the no-elite-sector map wherever the elite is active, and the stable steady state is strictly poorer, $k_{\text{elite}}^ < k_{\text{none}}^*$; (ii) a FOSD spread of the status-taste distribution lowers G pointwise (strictly where the elite is active) and strictly lowers the stable steady state; (iii) under positional status, both effects are strictly damped: $k_{\text{abs}}^* < k_{\text{pos}}^* < k_{\text{none}}^*$.*

One condition — the elite sector produces less output per unit of resources — now governs all three of the paper’s mechanisms: the static output decline (Proposition 2), the luxury property (Proposition 7), and the dynamic drag (Proposition 8). Together they compose the paper’s dynamic narrative: growth makes status affordable, status absorbs the resources that fund growth, and the economy converges to a steady state permanently poorer than its status-free counterfactual — with every generation optimizing, and every generation aware.

Two features of the dynamics deserve emphasis. First, there is no collapse: since the elite sector is unviable at low wealth (Proposition 7), the map near the origin is elite-free and the origin repels. Status spending impoverishes; it does not extinguish. The “self-destruction of the economy” takes the form of a permanently lower steady state, not a death spiral — and the mechanism that limits the damage is the same one that creates it: as wealth falls, the economy can no longer afford its elite. Second, positional dilution is again protective, now dynamically: because entry dilutes status, the elite mass responds far less to spreading taste, and steady-state wealth barely moves. Societies in which status is relative are — by that very relativity — partially immunized against the status drag.

On intertemporal welfare we claim less, deliberately. Within each period, the absolute-status allocation is Pareto optimal given k_t (Theorem 1); but future generations are not parties to any period’s core, and warm-glow bequests internalize the intergenerational margin only partially. Whether the equilibrium path is efficient among feasible paths — and what a generation-spanning planner could improve — is a question we flag for the sequel rather than resolve here.

8.4 Numerical continuation

Continuing the running example with $\gamma = 0.8$ (a 20% bequest share) and Θ calibrated so that the absolute-status steady state is exactly the static example’s wealth, $k^* = 2$: the steady state is stable ($G'(k^*) = 0.17$), with the static example’s prices, wages, cutoff, and elite mass reproduced as the steady-state allocation. The no-elite counterfactual steady state is 2.085; the positional steady state 2.045. As the taste scale rises from $s = 0.10$ to $s = 0.19$, the absolute-status steady state falls to 1.73 — a 16.8% wealth loss relative to

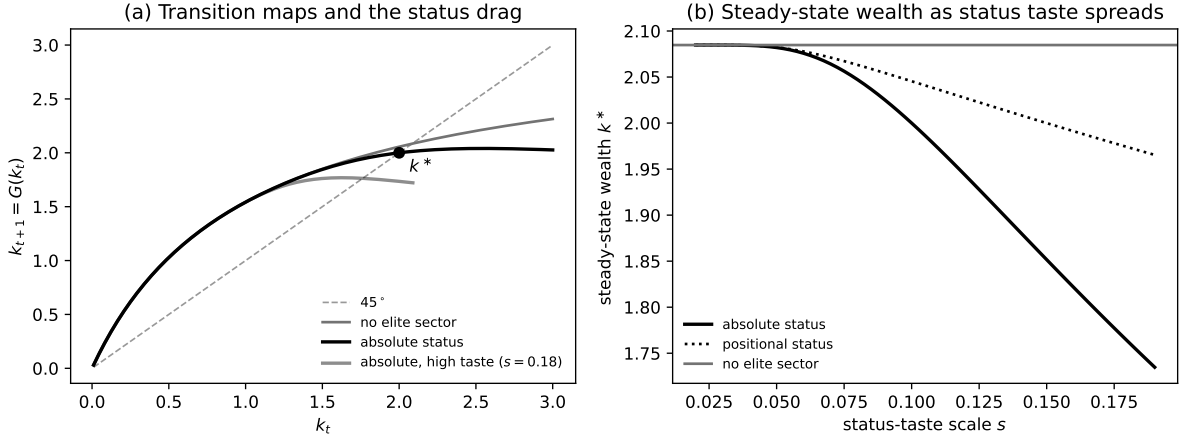


Figure 3: Dynamics. Panel (a): transition maps at baseline taste ($s = 0.1$) for the no-elite counterfactual and the absolute-status economy, and the absolute-status map after a taste spread ($s = 0.18$); the marked point is the calibrated steady state $k^* = 2$. Panel (b): stable steady-state wealth as the taste scale s rises — the absolute-status economy is impoverished by the spread of status taste, the positional economy is largely protected by dilution. Parameters as in Sections 6 and 7 with $\gamma = 0.8$, $\Theta = 5.141$.

the no-elite benchmark, with the elite sector swelling to about a fifth of the population — while the positional steady state falls only to 1.97 (a 5.7% loss). After a taste shift to $s = 0.18$, the transition $2.00 \rightarrow 1.74 \rightarrow 1.77 \rightarrow 1.76$ settles within three generations (the mild overshoot reflects the temporary output surge as displaced competent-sector resources are consumed). Figure 3 displays the maps and the steady-state decline. All computations are verified in `analysis/dynamic_model.py`.

9 Discussion and Planned Extensions

[Working notes; to be developed.]

GDP versus welfare. Propositions 1–2 and Theorem 1 jointly formalize a distinction the public debate conflates: what declines as elite status spreads is what statistical agencies measure, not what agents maximize. Any diagnosis of “inefficiency” from output or TFP data alone presumes status utility does not count — a normative stance that should be argued, not assumed.

Compensation as evidence. Corollary 1 makes elite compensation data informative rather than embarrassing. The model predicts exactly the observed pattern — conspicuous elite salaries, worse net positions after elite tuition — and identifies the premium as tuition reimbursement, not productivity. Two further channels would amplify observed

premiums without disturbing the mechanism: scarcity rents on the elite characteristic when its supply is capped (negative prices on a scarce characteristic, as in [Ellickson et al. 2006](#), Section 9.3.6), and heterogeneity in ability bundled with elite membership. The testable content is the decomposition itself: across elite labor markets, gross premiums should co-move with elite tuition, and estimated *net* returns to the elite career track should be at or below those of the non-elite track — consistent with the near-zero average selectivity premium of [Dale and Krueger \(2002\)](#).

Absolute versus positional status as an empirical question. The paper’s two halves assign opposite policy conclusions to observationally similar worlds — wasteful elite institutions persist in both — so the operative question is which preference structure obtains. The halves do differ observably: positional status implies self-damped responses of elite-sector size to demand shifts (Proposition 3(ii)) and predicts that historical expansions of elite institutions dilute the premium attached to them, while absolute status implies proportional pass-through. Developing this into a sharper empirical strategy is on the agenda for the framing phase.

Positional scarcity and rents. Positional status supplies the first of the amplification channels just noted: with v strictly decreasing, the elite characteristic is in endogenously limited supply, and extensions in which firms compete for a capped pool of incumbent elites would shift $q_{4.1}$ further toward scarcity rents — widening observed premiums beyond tuition reimbursement — without disturbing the marginal entrant’s status discount.

What remains. Three items are on the agenda for the next revision. First, dynasty-specific inheritance: uniform inheritance isolates the aggregate mechanism, but letting elite parents bequeath directly to their children would interact status persistence with wealth inequality in the manner of [Galor and Zeira \(1993\)](#), and would let the model speak to intergenerational mobility at the top. Second, parents who value their children’s *status* rather than the bequest itself — the natural sharpening of the warm-glow specification, which would make elite membership itself heritable in demand. Third, intertemporal welfare: each period’s absolute-status allocation is optimal given its inheritance, but the path’s efficiency across generations — and what, if anything, a generation-spanning planner could improve even in the absolute-status world — is open, and the answer plausibly differs between the absolute and positional regimes.

A Derivations and Proofs

All computations in this appendix are verified symbolically in `analysis/static_model.py`.

A.1 Demand and indirect utility

An agent with net income w solves $\max_{x_1, x_2} D\sqrt{x_1 x_2}$ subject to $x_1 + p_2 x_2 = w$, giving $x_1 = w/2$, $x_2 = w/(2p_2)$, and indirect utility $D w/(2\sqrt{p_2})$.

A.2 Dominated lists

Lemma 2. *At any prices satisfying (4) with $r, \tau r > 0$, every feasible list other than the five in Section 4 is strictly dominated: (i) schooling with no job yields $D \leq \delta_s < 1$ and net income $k - q_j < k$, dominated by leisure; (ii) schooling plus a laborer job multiplies the laborer's D by $\delta_s < 1$ and subtracts tuition, dominated by the same laborer job alone; (iii) double schooling adds a factor δ_s and tuition, dominated by the single required school; (iv) an elite-school graduate cannot hold membership $m_{3,1}$ nor a serious-school graduate $m_{4,1}$ (consumption sets), so no other manager lists are feasible.* $\square$

A.3 Proof of Lemma 1

Budget balance. Schools: $q_1 - r = 0$, $q_2 - \tau r = 0$ by (4). Competent firms: $q_{3,1} + 2q_{3,2} - c + p_2 \phi = (k - r - \mu k) + 2(k - \lambda k) - c + p_2 \phi = 0$ by (5). Elite firms: $q_{4,1} + 2q_{4,2} - \beta c + p_2 \alpha \phi = 0$ reduces, using (5), to $w_E = Z$, which is (7). *Optimization.* By Lemma 2 only the five vocational choices need comparison; by construction of (2) and (3), types below η^* are indifferent among the four non-elite choices and strictly prefer them to the elite path, types above strictly prefer the elite path, and the marginal type is indifferent. *Consistency and material balance.* The stated measures fill every group type in the required proportions; market clearing (8) solves

$$n = \frac{k - (1 - 3m^*)\frac{k}{2} - m^* \left[\frac{w_E}{2} + w_{L^*} + \tau r + \beta c \right]}{\frac{w_M}{2} + w_L - \frac{3k}{2} + r + c}, \quad (18)$$

positive under Assumption 1. *Walras.* Substituting (18) and (7) into the y_2 market, demand $\sum (\text{measures}) \times w/(2p_2)$ equals supply $\phi(n + \alpha m^*)$ identically (machine-verified). Uniqueness of the interior configuration follows because (2)–(7) are implied by indifference and budget balance whenever all memberships are active, and (18) is linear in n . $\square$

A.4 Proof of Proposition 1

Subtracting the two firm budget-balance conditions, $p_2\phi(1 - \alpha) = (w_M - w_E) - 2(w_{L^*} - w_L) - (\beta - 1)c - (\tau - 1)r$ after substituting (4); the left side is positive iff $\alpha < 1$, given $p_2 > 0$. $\square$

A.5 Proof of Proposition 2

η^* solves (7), which contains no object depending on F ; a FOSD shift weakly raises $1 - F(\eta)$ pointwise, strictly at η^* by the strict-increase assumption on F , so m^* rises. Differentiating $Y = \phi(n + \alpha m^*)$ with n from (18),

$$\frac{dY}{dm^*} = \phi \left(\alpha - 1 + \frac{d(n + m^*)}{dm^*} \right), \quad \text{sign } \frac{dY}{dm^*} = \text{sign} [\alpha(c + r) - (\beta c + \tau r)],$$

where the second equality is by direct computation (machine-verified): the denominator of (18) equals $\frac{1}{2}[2c + 2r + (\mu + 2\lambda - 3)k] > 0$ and the numerator of $\alpha - 1 + d(n + m^*)/dm^*$ reduces to $\alpha(c + r) - (\beta c + \tau r)$ over that denominator. $\square$

B Verification of Framework Assumptions

Consumption sets. $X_a = \mathbb{R}_+^2 \times \mathbf{Lists}_a$ with $\mathbf{Lists}_a$ the lists satisfying the schooling prerequisites and the at-most-one-school, at-most-one-job bounds; the membership bound is $M = 2$. Free upward disposal in private goods holds. The correspondence $a \mapsto X_a$ is constant in a (types differ only through u), hence measurable.

Utility. $u(x, \ell; \eta)$ is continuous and strictly increasing in x for each ℓ , and jointly measurable since $\eta \mapsto D(\ell; \eta)$ is continuous and $a \mapsto \eta_a$ measurable.

Desirable endowments. $u_a(e_a, 0) = h(k, 0) = 0$. If $u_a(x, \ell) > 0$ then $x \gg 0$, so any reduction $x' \leq x$, $x' \neq x$, remains in X_a (consumption sets impose no lower bound above zero); hence endowments are desirable in the sense of Ellickson et al. (2006, Section 9.4), and their Theorem 1 delivers membership of every equilibrium state in the strong core and strong Pareto optimality.

Existence. Not invoked: Lemma 1 exhibits equilibrium in closed form. (For non-interior parameter configurations, existence follows from Theorem 2 of Ellickson et al. 2006 upon verifying group irreducibility; this is routine but omitted until needed.)

C Positional Extension: Proofs

All computations are verified in `analysis/positional_model.py`.

C.1 Proof of Proposition 3

The occupational-indifference conditions for non-elite choices do not involve M , so (2) and (5) are unchanged, as is the elite break-even income $w_E = Z$ from firm budget balance. The cutoff type solves $(1 + \eta v(M)) \delta_s \delta_m w_E = k$, giving (13); sorting is strict above and below the cutoff because (12) is strictly increasing in η for fixed M . Consistency of M with sorting is (14). Since v is continuous and strictly decreasing and F is continuous and strictly increasing, Φ is continuous and strictly decreasing, with $\Phi(0) = 1 - F(\eta_0^*) < 1$ and $\Phi(1) \geq 0$; a unique fixed point $M^* \in [0, 1)$ exists by the intermediate value theorem, and $M^* > 0$ whenever F has unbounded support. (i) Since $v(M^*) < 1$ for $M^* > 0$, $\eta^* = \eta_0^*/v(M^*) > \eta_0^*$, so $M^* = 1 - F(\eta^*) < 1 - F(\eta_0^*) = M_0$. (ii) A FOSD shift raises $1 - F(\cdot)$ pointwise, shifting Φ up; with Φ strictly decreasing, the fixed point rises, but the induced fall in v raises the cutoff, damping the increase: formally $dM^* = d\Phi/(1 + |\partial\Phi/\partial M|) < d\Phi = dM_0$. (iii) Immediate, since given M^* the remaining conditions coincide with those of Lemma 1. $\square$

C.2 Proof of Proposition 4

Work at the equilibrium of Proposition 3. Shutting one elite triple and moving its three members to leisure at their reservation utility changes aggregate excess demand by (A_e, B_e) in (input, output):

$$A_e = \tau r + \beta c + \frac{w_E}{2} + w_{L^*} - \frac{3k}{2}, \quad B_e = -\alpha \phi + \frac{w_E + 2w_{L^*} - 3k}{2p_2},$$

and elite-firm budget balance implies $A_e + p_2 B_e = 0$. Opening one competent triple staffed from the (indifferent) leisure pool changes excess demand by $(-A_c, B_c)$ with $A_c = c + r + (w_M + 2w_L - 3k)/2$, $B_c = \phi - (w_M + 2w_L - 3k)/(2p_2)$, and competent budget balance implies $A_c = p_2 B_c$. Both vectors are orthogonal to $(1, p_2)$, hence collinear; shutting mass $dM > 0$ of elite triples and opening $dn = (A_e/A_c) dM$ competent triples restores material balance in both goods *exactly* (machine-verified). All movers and all non-elite agents are at reservation utility throughout, so no agent is worse off. The elite mass falls to $M^* - dM$, so v rises, and every inframarginal elite's utility rises by $\eta [v(M^* - dM) - v(M^*)] \delta_s \delta_m w_E h$ -units > 0 . Finally, transfer an arbitrarily small amount of each good from the (strictly

better off) elites to all agents pro rata: by continuity the elites remain strictly better off, and strict monotonicity makes almost every agent strictly better off. The resulting state is feasible and strictly Pareto dominates the equilibrium state; the grand coalition can implement it, so the state is blocked and not in the core. $\square$

C.3 Planner cutoff condition, the wedge (15), and Proposition 5

Consider the one-parameter family of feasible states indexed by the cutoff $\hat{\eta}$, constructed as in the proof of Proposition 4 (members outside the elite held at reservation utility; material balance restored via the competent margin). Elite utility is $U(\eta, \hat{\eta}) = (1 + \eta v(M(\hat{\eta}))) \delta_s \delta_m w_E H$ with $M(\hat{\eta}) = 1 - F(\hat{\eta})$ and $H \equiv 1/(2\sqrt{p_2})$. Differentiating aggregate elite utility and using $M'(\hat{\eta}) = -f(\hat{\eta})$:

$$\frac{d}{d\hat{\eta}} \int_{\hat{\eta}}^{\infty} U dF = - \underbrace{U(\hat{\eta}, \hat{\eta}) f(\hat{\eta})}_{\text{marginal type's utility}} + \underbrace{f(\hat{\eta}) |v'(M)| \delta_s \delta_m w_E H \int_{\hat{\eta}}^{\infty} \eta dF}_{\text{de-dilution of inframarginals}}.$$

At an interior optimum the marginal type's *surplus* over reservation utility equals the second term divided by $f(\hat{\eta})$; converting to income units by dividing by the marginal type's marginal utility of income $(1 + \hat{\eta}v)\delta_s \delta_m H$ gives the wedge (15). At laissez-faire the marginal surplus is zero while $\Omega > 0$: over-entry. For Proposition 5: (i) with tax t and rebate $T = tM$, the entrant's income becomes $w_E = Z - t$ and the fixed-point map remains continuous and strictly decreasing, so uniqueness persists and $M(t)$ is strictly decreasing for small t (machine-verified in the example, monotone by the same argument as Proposition 3); (ii) at $t = 0$ every private margin is at a first-order optimum, so the derivative of utilitarian welfare reduces to the uninternalized de-dilution term, which is strictly positive; equating the marginal entrant's income loss to the wedge gives $t^* = \Omega$; (iii) $dM/dt < 0$ and condition (11) give $dY/dt > 0$ by Proposition 2. The global welfare-maximizing tax in the numerical example, which accounts for all equilibrium feedback including the rebate, is computed in `analysis/positional_model.py`. $\square$

D Dynamics: Proofs

All computations are verified in `analysis/dynamic_model.py`.

Lemma 3 (Period equivalence). *Under preferences (16), an agent with net income w facing (p_2) chooses $x_1 = (\gamma/2)w$, $p_2 x_2 = (\gamma/2)w$, $p_2 b = (1 - \gamma)w$, and attains indirect utility*

$D w \Psi(p_2)$ with $\Psi(p_2) = (\gamma/2)^\gamma (1 - \gamma)^{1-\gamma} p_2^{-(\gamma/2+1-\gamma)}$. Hence all occupational-indifference conditions, membership prices (4), the output price (5), the cutoff (7), and the positional fixed point (14) hold at wealth k_t exactly as in the static sections. Market clearing in the input good reads $(\gamma/2)W + n(r+c) + m^*(\tau r + \beta c) = k$, which is linear in n ; output clearing holds identically given it (Walras, machine-verified), and at $\gamma = 1$ the static condition (8) is recovered. $\square$

D.1 Proof of Proposition 6

(i) is Lemma 3. (ii): aggregate income satisfies $W = k + n(\mu + 2\lambda - 3)k + m^*e$ with $e \equiv w_E + (2\lambda^e - 3)k$; substituting the clearing solution for n gives $W(k)$ in closed form, and $G = \Theta(1 - \gamma)\phi W/\Pi$ with $\Pi(k) = c + r + (\mu + 2\lambda - 3)k$. As $k \rightarrow 0^+$ the elite sector is inactive (Proposition 7) and $W \rightarrow k$, so $G/k \rightarrow \Theta(1 - \gamma)\phi/(c+r)$; as $k \rightarrow \infty$, n and W/Π converge, so G is bounded (limits machine-verified). (iii): with $G(0) = 0$, $G'(0^+) > 1$, G continuous and bounded, G crosses the diagonal from above at least once; at any such crossing the steady state is stable if $|G'| < 1$, verified in the example ($G' = 0.31$). The no-forecasting claim is immediate: (16) depends on own consumption and own gift only, and period- t prices depend only on k_t . $\square$

D.2 Proof of Proposition 7

$\eta_0^*(k) = \mu k/Z(k) - 1$ with $Z = \alpha\Pi - \beta c - \tau r + (3 - 2\lambda^e)k$. Then $d\eta_0^*/dk = \mu(Z - kZ')/Z^2$ and direct computation (machine-verified) gives $Z - kZ' = \alpha(c+r) - \beta c - \tau r$. Hence η_0^* is strictly decreasing iff $\alpha(c+r) < \beta c + \tau r$, and $m^* = 1 - F(\eta_0^*)$ (or its positional analogue, whose fixed-point map shifts up as η_0^* falls) is strictly increasing wherever $0 < Z < \mu k$. Viability: $Z(k) \leq 0$ for k below the root of Z , where no cutoff exists and the elite sector is empty. $\square$

D.3 Proof of Proposition 8

Note $e = w_E + (2\lambda^e - 3)k = \alpha\Pi - \beta c - \tau r$: the elite triple's value added. Substituting the clearing solution for n into W and differentiating at fixed k ,

$$\text{sign } \frac{\partial W}{\partial m^*} = \text{sign } \left[(r+c)e - (\tau r + \beta c)(\Pi - r - c) \right] = \text{sign } \left[\Pi(\alpha(c+r) - \beta c - \tau r) \right],$$

(machine-verified), and $\partial G/\partial m^*$ has the same sign since Π is fixed given k . Under (11) the sign is negative: (i) follows pointwise and hence for the stable steady state (G falls,

the crossing with the diagonal moves left); (ii) a FOSD shift raises $m^*(k)$ at every k with the elite active (Propositions 2 and 3(ii)), lowering G pointwise; (iii) under positional status the same shift raises m^* by strictly less (Proposition 3(ii)), so G falls by strictly less at every k , and the steady states order as stated (verified in the example). $\square$

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